

Data Center Capacity Evolution

Data center capacity is commonly measured in installed IT power (megawatts, MW) or in server count. Both have grown dramatically:

Era	Dominant driver	Typical facility	Scale
1960s–1980s	Mainframe computing	Corporate or government machine room	10–100 kW; one to a few machines
1990s	Client/server; early web hosting	Purpose-built server room	100 kW–1 MW
2000s	Dot-com boom; search engines; early SaaS	Colocation and early hyperscale	1–10 MW per facility
2010s	Cloud computing; mobile; video streaming	AWS, Azure, GCP mega-campuses	10–200 MW per campus
2020–2024	Pandemic-driven cloud migration; early AI	Hyperscale campuses + GPU clusters	100–500 MW per campus
2025+	Large-scale AI training and inference	AI-optimised gigawatt campuses	500 MW–1+ GW per campus

Global installed data center capacity reached approximately 60–70 GW of IT load by 2024, having roughly doubled every four years through the cloud era. The AI acceleration beginning in 2022–2023 is compressing that doubling interval: industry analysts project 200+ GW of global capacity by 2030 if announced construction projects proceed on schedule.

Water consumption has grown alongside power draw. A 100 MW facility may consume 1–3 million gallons of water per day for evaporative cooling — a significant concern in water-scarce regions, driving adoption of dry cooling and liquid immersion alternatives.